

EXPERIMENT-6

Name: Siddhant Mehta

UID: 23BIS70162

Title

Interactive SVG Drawing Tool with Mouse Event Handlers

Objective

Design and build a web-based drawing tool using SVG where users can draw shapes interactively using their mouse. This task deepens your understanding of JavaScript DOM manipulation, SVG element creation, and advanced event handling.

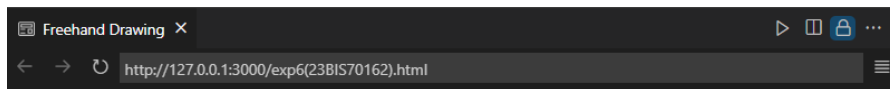
Task Description

Create a web page that includes an SVG area acting as a drawing canvas. When the user clicks and drags inside this area, a shape (for example, a line or rectangle) should be dynamically drawn following the mouse movement. The tool should respond to mouse events such as mousedown, mousemove, and mouseup to create and update the shapes in real time. Users should be able to draw multiple shapes by repeating the click-and-drag action. All drawing interactions must be implemented using JavaScript without reloading the page.

INPUT SCREENSHOTS:

```
exp6(23BIS70162).html > html > body > script
1  <!DOCTYPE html>
2  <html>
3  <head>
4  | <title>Freehand Drawing</title>
5  </head>
6  <body>
7  | <h2>Freehand Drawing in SVG</h2>
8  | <p id="pos">X:0, Y:0</p>
9
10 | <svg id="mysvg" width="600" height="400" style="border:1px solid black;">
11 | </svg>
12
13 <script>
14 |   var svg = document.getElementById("mysvg");
15 |   var pos = document.getElementById("pos");
16 |   var drawing = false;
17 |   var path;
18 |   var points = "";
19
20 |   svg.addEventListener("mousedown", function(e) {
21 |     drawing = true;
22 |     points = e.offsetX + "," + e.offsetY;
23 |     path = document.createElementNS("http://www.w3.org/2000/svg", "polyline");
24 |     path.setAttribute("points", points);
25 |     path.setAttribute("stroke", "black");
26 |     path.setAttribute("fill", "none");
27 |     svg.appendChild(path);
28 |   });
29
30 |   svg.addEventListener("mousemove", function(e) {
31 |     pos.innerText = "X:" + e.offsetX + ", Y:" + e.offsetY;
32 |     if (drawing) {
33 |       points += " " + e.offsetX + "," + e.offsetY;
34 |       path.setAttribute("points", points);
35 |     }
36 |   });
37
38 |   svg.addEventListener("mouseup", function(e) {
39 |     drawing = false;
40 |   });
41 </script>
42 </body>
43 </html>
44
```

OUTPUT SCREENSHOTS:



Freehand Drawing in SVG

X:145, Y:19



Please reopen the preview.

